

Data representation test - Answer Key

1. What are the two broad categories of cells?

Answer: B

Explanation: The textbook states that the two broad categories of cells are prokaryotic and eukaryotic.

2. What is the main function of the cell membrane?

Answer: To control what enters and exits the cell

Explanation: The cell membrane is a selectively permeable barrier that regulates the movement of substances in and out of the cell.

3. The nucleus is the control center of the prokaryotic cell.

Answer: False

Explanation: The nucleus is the control center of the eukaryotic cell, not the prokaryotic cell.

4. What is the process by which cells convert nutrients into usable energy?

Answer: B

Explanation: The textbook states that cellular respiration is the process by which cells convert nutrients into usable energy in the form of ATP.

5. What are the three main stages of cellular respiration?

Answer: Glycolysis, the Krebs cycle, and the electron transport chain

Explanation: The textbook lists these three stages as the main stages of cellular respiration.

6. Glycolysis is an aerobic process.

Answer: False

Explanation: The textbook states that glycolysis is an anaerobic process, meaning it does not require oxygen.

7. What is the purpose of the electron transport chain?

Answer: C

Explanation: The electron transport chain produces a large majority of the cell's ATP through the process of chemiosmosis.

8. What are the two processes by which eukaryotic cells can divide?

Answer: Mitosis and meiosis

Explanation: The textbook states that eukaryotic cells can divide through either mitosis or meiosis.